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Live Event Detection for People's Safety Using NLP and Deep Learning

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INTRODUCTION

- Today, humans pose the greatest threat to society by getting involved in robbery, assault, or homicide activities.
- Such circumstances threaten the people working alone at night in remote areas especially women.
- Any such kind of threat in real time is always associated with a sound/noise which may be used for an early detection.
- Numerous existing measures are available but none of them sounds efficient due to lack of accuracy, delays in exact prediction of threat.

INTRODUCTION

- The solution for this problem can be attained with the help of sound based NLP(Natural Language Processing) techniques.
- It mainly uses two modules -Live recording module and Prediction module.
- The live recording module continuously listen for any sound from person's surroundings ,and send the recorded sound to the prediction module.
- The prediction module analyzes the sound is danger or not.
- If it is danger ,then an automatic and immediate alert given to emergency services

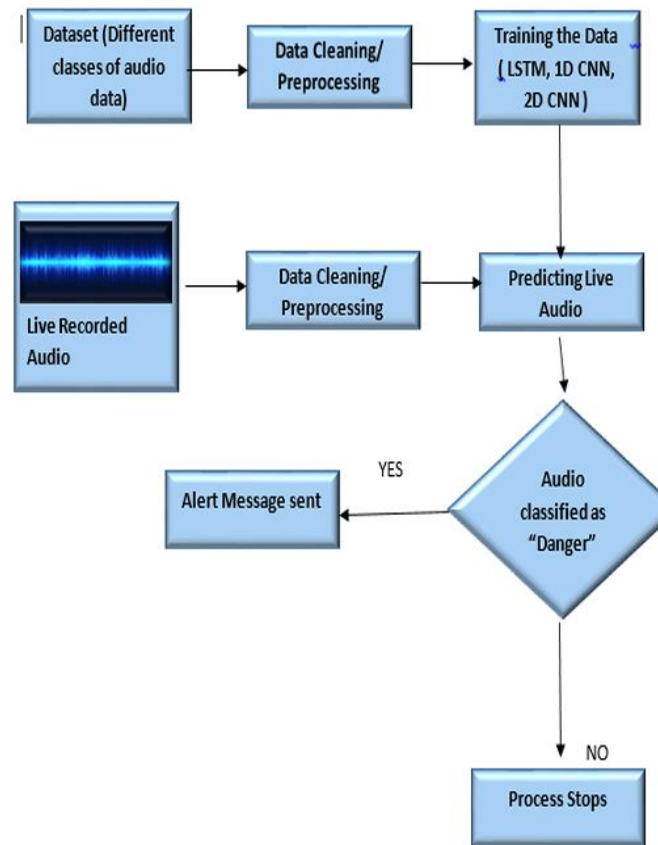
INTRODUCTION

- To carry out accurate prediction, training and testing of Kaggle dataset is carried out using 3 deep learning models namely LSTM, 1D CNN , 2D CNN.
- The key advantage of this system is that it does not require any special external hardware/wearable devices.
- It can be implemented with the help of a mere smartphone configured accordingly.
- So, a person would not have to carry bulky hardware around with them, but only remember to carry their phones with them.

NATURAL LANGUAGE PROCESSING

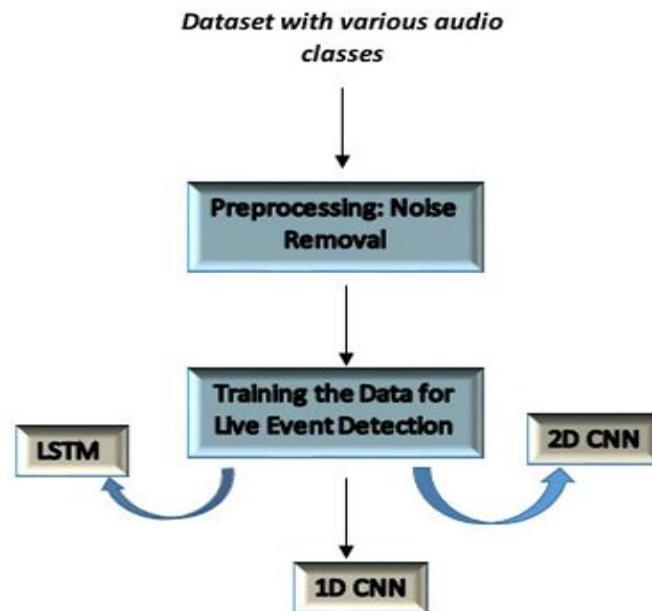
- Natural Language Processing (NLP) is a field of artificial intelligence (AI) that focuses on the interaction between computers and humans through natural language.
- The goal of NLP is to enable computers to understand, interpret, and generate human language in a way that is both meaningful and useful.
- Main applications of NLP are:
 - Virtual Assistants
 - Chatbots
 - Text Auto Correction and Prediction
 - Sentiment Analysis
 - Voice-to-Text

ARCHITECTURE



DATASET

- The dataset described here is a customized dataset combining parts of three different audio datasets -UrbanSound8K ,Environmental sounds,Scream dataset.
- In this dataset 13 classes of audio signals to be trained on the three deep learning models.



EXPLORATORY DATA ANALYTICS (EDA)

- Exploratory Data Analysis (EDA) in audio signal processing involves analyzing and visualizing audio data to understand its characteristics ,detect anomalies and uncover patterns
- It is applied on audio data before developing or applying machine learning or deep learning models.

The major steps in EDA are:

- **Sampling**
- **Converting Time domain into Frequency domain**
- **Mel Spectrogram**
- **Cleaning the Sound Data**

EXPLORATORY DATA ANALYTICS (EDA)

1. Sampling:

Converting a continuous audio signal into a discrete signal

It involves measuring the amplitude of the audio signal at regular intervals

Common sampling rate: 44.1KHz ,48 kHz

Nyquist Theorem: $f_s \geq 2 f_{max}$

f_s :sampling rate.

f_{max} : maximum frequency component in the signal.

2. Converting Time domain into Frequency domain

- Fast Fourier Transform (FFT)
- Short-Time Fourier Transform (STFT)

EXPLORATORY DATA ANALYTICS (EDA)

- **Fast Fourier Transform (FFT)**

FFT provides a frequency-domain representation of an entire signal and is useful for global analysis of frequency content.

It is best suited for stationary (periodic) signals.

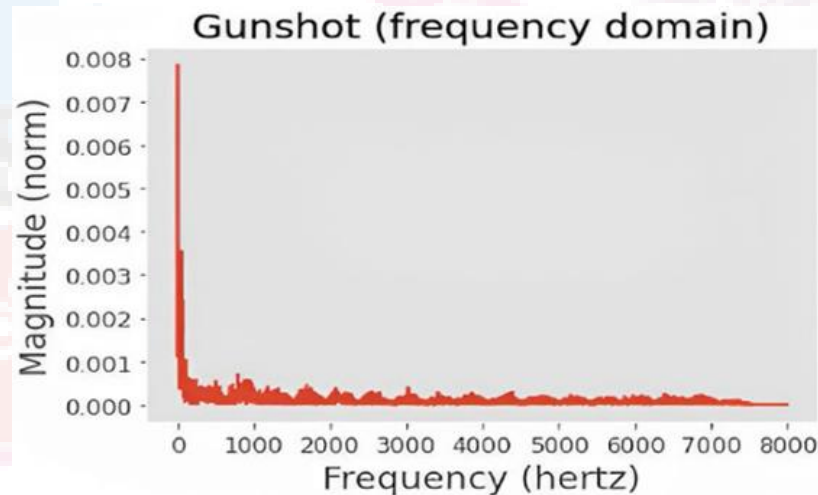
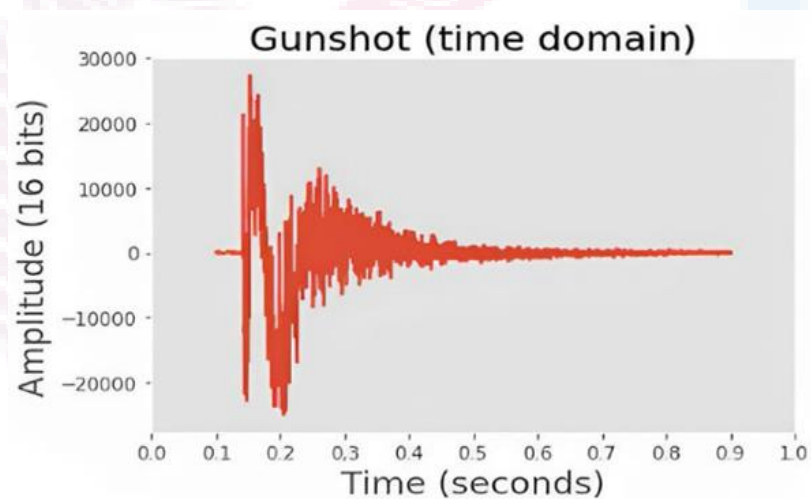
- **Short-Time Fourier Transform (STFT)**

It provides both time and frequency domains of short segments of a signal

It is best suited for non periodic signals.

Shows how frequency components change over time

EXPLORATORY DATA ANALYTICS (EDA)



EXPLORATORY DATA ANALYTICS (EDA)

3. Mel Spectrogram

- A Mel Spectrogram is a time-frequency representation of an audio signal that has been transformed to the Mel scale, which is a perceptual scale designed to match human hearing.
- It is generated by passing Mel Filterbank

Mel Scale

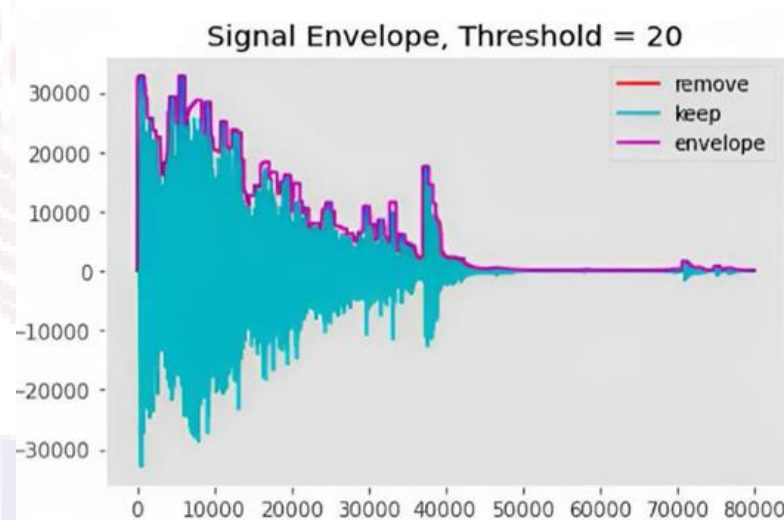
The Mel scale is a perceptual scale of pitches that is designed to reflect how humans perceive differences in pitch or frequency.

The Mel scale is nonlinear; it compresses higher frequencies and expands lower frequencies, reflecting how humans perceive pitch

EXPLORATORY DATA ANALYTICS (EDA)

4. Cleaning the Sound Data

The silent zones or dead spaces in the audio can be removed by creating a customized signal envelope



Down sampling:
sampling rate:16KHz
converts into single mono channel

DEEP LEARNING MODELS

1 Dimensional CNN (Convolution Neural Network)

A 1D-CNN for audio processing takes raw audio data, applies convolutional filters over time, and extracts features that are used to classify the audio into categories.

There are mainly 6 layers in 1D-CNN:

- 1. Convolutional Layers:**

To extract features from the audio signal by sliding filters over the time dimension.

- 2. Time Distributed Layers:**

To apply the same convolutional operation over different segments of the time series.

1. 1-Dimensional CNN (Convolution Neural Network)

3. **GlobalMaxPooling Layer:**

To take the maximum value from the features, which simplifies the model and focuses on the most prominent feature in each set.

4. **Dropout Layer:**

To randomly deactivate some neurons during training, which helps prevent overfitting.

1. 1-Dimensional CNN (Convolution Neural Network)

5. **Final Dense Layers and Classification:**

The final part of the network uses dense (fully connected) layers to classify the features extracted from the audio data.

6. **Output Layer:**

The output is passed through a SoftMax activation function, which converts the outputs into probabilities for different classes (e.g., different sound categories).

2. 2-Dimensional CNN (Convolution Neural Network)

- The 2D-CNN for audio processing takes on 2D representations of audio like Mel spectrograms.
- Convolutional filters move across both time and frequency dimensions, capturing spatial patterns in the spectrogram.
- More effective for capturing complex patterns in the time-frequency domain compared to 1D-CNNs.

The major phases in 2D-CNN are:

1. **2D Convolutional Layers**

To detect and learn spatial features (specific frequency bands, temporal patterns, or textures) from the 2D input data

2. 2-Dimensional CNN (Convolution Neural Network)

2. **Activation Functions:**

ReLU (Rectified Linear Unit): To learn complex patterns

Hyperbolic Tangent (tanh): Used in the first layer

3. **MaxPooling Layers:**

Reduce computational complexity and preventing overfitting

4. **Flattening Layer:**

Converts the final 2D feature map into a 1D vector.

2. 2-Dimensional CNN (Convolution Neural Network)

5. **Fully Connected Layer:**

The final part of the network uses fully connected layers to classify the features extracted from the audio data.

6. **Output Layer:**

The output is passed through a SoftMax activation function, which converts the outputs into probabilities for different classes (e.g., different sound categories).

3. LSTM (Long Short-Term Memory) Model

- LSTM networks are designed to handle and analyze sequences of data, such as audio, video, or time series.
- They can remember patterns and relationships over time, making them suitable for tasks like speech recognition, translation, and more.

Key Components:

- Memory Cell
- Gates

Layers in LSTM

- Time Distributed Dense Layer
- Bidirectional LSTM Layer
- Final dense layers

3. LSTM (Long Short-Term Memory) Model

- **Time Distributed Dense Layer:**

This layer initially processes features to learn and reduce them before feeding them into the LSTM layers.

- **Bidirectional LSTM Layer:**

This layer processes data in both forward and backward directions to capture patterns from both sides of the sequence.

Further Processing:

- Dense Layers
- MaxPooling
- Flattening

3. LSTM (Long Short-Term Memory) Model

- **Final Dense Layers and Classification:**

The final part of the network uses dense (fully connected) layers to classify the features extracted from the audio data.

- **Output Layer:**

The final layer uses SoftMax to output probabilities for each class, providing the model's predictions.

LIVE AUDIO RECORDING AND PREDICTION MODULE

1. Audio Recording:

Live Recording: Captures live audio from the environment.

Automatic Stopping: The recording stops automatically when no more noise is detected.

2. Pre-Processing Steps:

Trimming Silence: Remove Silence

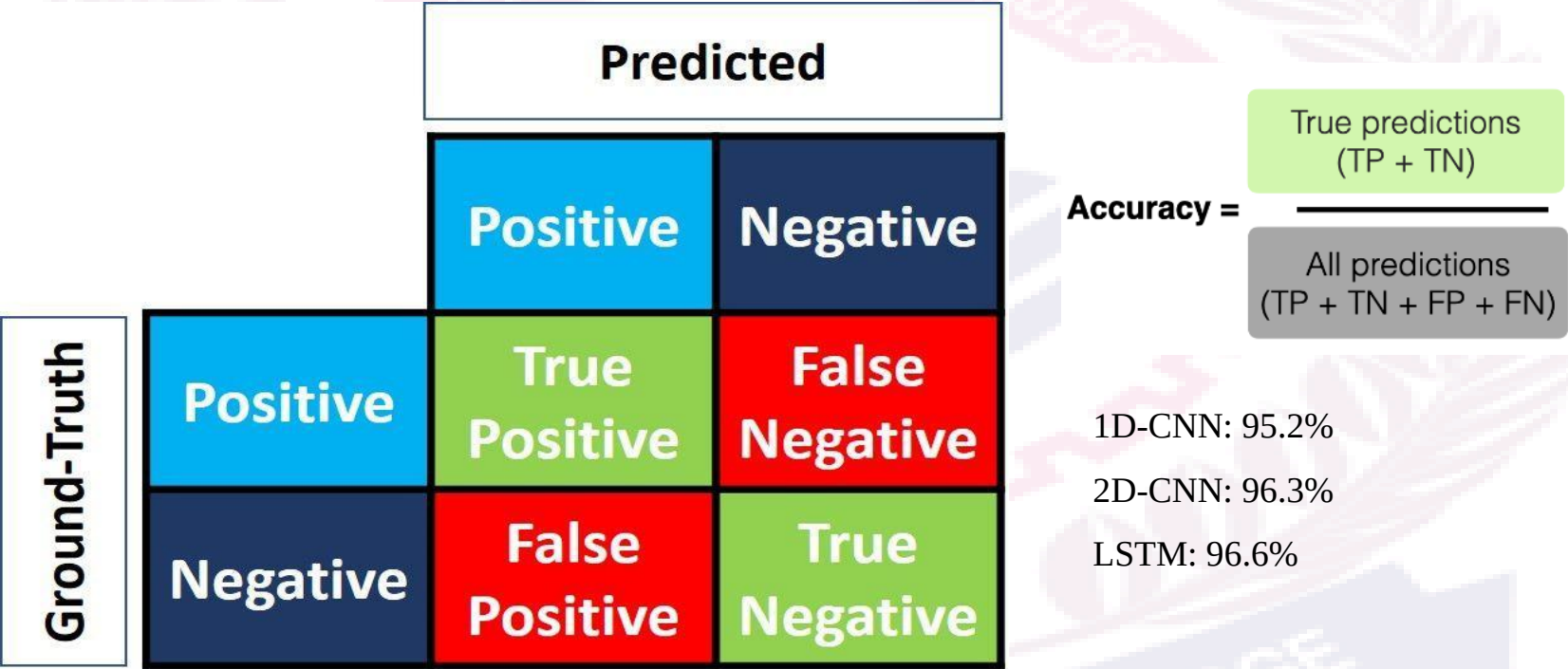
Normalization: Volume Adjustment

Audio Padding: Padding Addition

Cont..

3. **Saving Processed Audio:** Saves the pre-processed audio as a .wav file
4. **Automatic Prediction Module Activation:** Uses one of the pre-built deep learning models to classify the recorded audio.
5. **Automatic Alert System:** Automatic alert message is immediately sent to the registered contact
It also contains the audio file saved from the live recording

CONFUSION MATRIX



CONCLUSION

- The research successfully developed a software-based threat identification system that can detect dangerous situations based on ambient noise and provide immediate assistance by notifying emergency contacts.
- The system uses three deep learning models with an average accuracy of 96.03%.
- IT offers an effective solution for real-time threat detection and alerting and demonstrates the potential of deep learning in practical applications like audio analysis.

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THANK YOU